

Adversarial Robustness of U-Net for Inland Water Body Segmentation from SAR Images

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ABSTRACT

Inland water body segmentation from Synthetic Aperture Radar (SAR) images is crucial for applications such as flood monitoring and environmental assessment. While convolutional neural networks (CNNs), particularly U-Net, achieve state-of-the-art performance in SAR segmentation, they depend heavily on manually annotated masks for training. These annotations are prone to label noise, especially around complex river geometries, which can propagate to the models. In this work, we simulate annotation errors using morphological label noise—generated via controlled erosion and dilation, and evaluate the robustness of the U-Net under such data poisoning attacks. Our findings show that U-Net can tolerate a limited degree of corruption before performance degrades sharply, underscoring the importance of annotation quality. We also release the dataset, code, and adversarial examples to facilitate robust model development. (GitHub link: <https://github.com/GVCL/IWSeg-SAR-Poison.git>)

KEYWORDS

Deep learning, SAR Images, Image Segmentation, U-Net, Label Noise, Adversarial Attacks, Morphological Operations

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1 INTRODUCTION

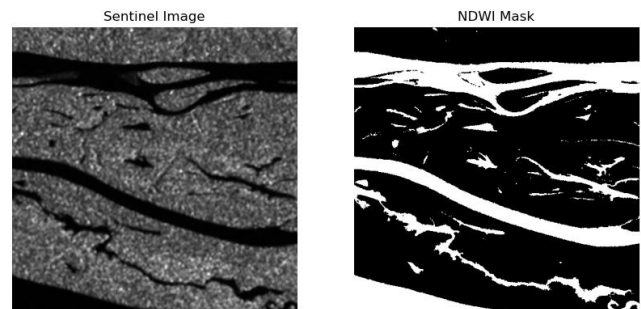


Figure 1: Example of Sentinel-1 SAR images of the Amazon River Basin (left) in our dataset, along with their generated binary water masks (right).

Water body segmentation is vital in remote sensing, supporting flood mapping [7, 9, 12], glacial lake monitoring [13], sedimentation studies [8], and environmental assessments. Synthetic Aperture Radar (SAR) offers all-weather, high-resolution data, yet inland water segmentation remains difficult due to the complex geometry of river basins and tributaries.

Deep learning methods, particularly encoder-decoder CNNs like U-Net [11], surpass traditional approaches but depend heavily on annotated masks, which often contain errors from boundary ambiguities, width variations, and mislabeling [2, 14]. Such errors can impair training, motivating us to study their impact through adversarial robustness.

Unlike typical adversarial attacks based on imperceptible perturbations [1, 10], we introduce *morphological label noise*—annotation corruptions via erosion and dilation—as a proxy for human errors in complex river geometries (e.g., the Amazon Basin). We evaluate U-Net under varying corruption levels to assess its tolerance to noisy supervision.

Our main contributions are:

- (1) A framework to simulate annotation errors with morphological operations, and a dataset with adversarial examples. Our novel benchmark dataset is available online [5].
- (2) An analysis of U-Net’s robustness across corruption levels.

2 METHOD

We generated our dataset using Sentinel-1 SAR imagery for the Amazon Basin where the study area has significant geomorphological diversity [4]), thus making it a challenging dataset. The ground truth water masks are derived from NDWI thresholding of corresponding Sentinel-2 MSI data. An example SAR image and mask is shown in Figure 1. The detailed data preparation process involves the standard SAR image preprocessing [6].

Erosion and dilation are morphological operations commonly used in image processing, which we apply to generate controlled corruption of annotated masks. Since human annotators may introduce errors at land–water boundaries, especially in complex hydrological geometries, these operations are well-suited as they add or remove pixels at class boundaries. To simulate real-world non-uniformity, we introduce randomness in the kernel sizes during corruption. We apply corruption at levels {2%, 12%, 15%, 17%, 20%, 25%, 30%}. The resulting corrupted masks are then used in adversarial experiments to assess model robustness against boundary errors, which appear as inadvertent ones made by humans.

- (1) Out of the train and the validation images, we select one-third of the images for corruption using erosion, one-third using dilation, and one-third is left uncorrupted.
- (2) For an experiment, we then set a threshold on the number of corrupted pixels allowed.
- (3) Erosion and dilation are applied to the corresponding masks iteratively, till either 100 iterations are over, or the corruption crosses the threshold, following which we choose the previous highest corrupted version.

Although we set limits on the number of corrupted pixels, these are not always reached in practice, particularly for erosion. Since erosion reduces white pixels, it saturates once an insufficient number remain, preventing further corruption. This effect is illustrated in Figure 2, which shows box plots of corrupted pixels for erosion and dilation across all training images. For erosion, the number of corrupted pixels stagnates beyond 20%, with median and lower quartile values remaining almost unchanged, while dilation follows the expected increasing trend.

3 EXPERIMENTS & RESULTS

Table 1 presents the evaluation metrics of the U-Net model across different corruption levels. The model achieves around 95% segmentation accuracy, with performance largely unaffected up to 20% corruption and only moderate degradation beyond 25%. Figure 3 further illustrates this effect: as erosion increases, the proportion of white pixels decreases, yet the U-Net output, though blurred before thresholding, still captures the river geometry accurately, confirming the model’s robustness to substantial corruption. Thresholding restores clear masks until 20% corruption, but beyond that, the model predictions are poor (30%).

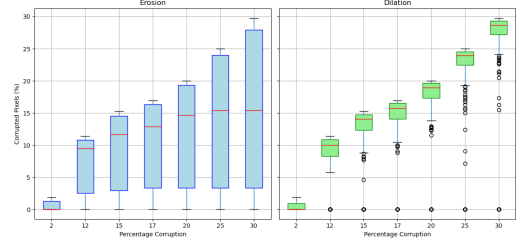


Figure 2: A box plot showing the range of actual corruption in the training data for different corruption levels.

Table 1: Performance of the baseline U-Net model on our dataset at various corruption levels

Corruption Level	Dice Coeff.	IoU	Precision	Recall	F1 Score	Specificity	Accuracy
0%	0.770	0.823	0.896	0.869	0.865	0.971	0.955
2%	0.775	0.825	0.896	0.873	0.868	0.968	0.955
12%	0.769	0.822	0.899	0.864	0.865	0.975	0.955
15%	0.768	0.819	0.890	0.866	0.863	0.971	0.954
17%	0.758	0.814	0.903	0.854	0.859	0.979	0.952
20%	0.725	0.790	0.902	0.830	0.841	0.981	0.946
25%	0.597	0.708	0.875	0.759	0.768	0.984	0.920
30%	0.253	0.496	0.818	0.588	0.567	0.996	0.828

We selected U-Net for adversarial experimentation due to its strong performance with smaller training sets, faster training, and stability to annotation errors [3]. Future work includes extending experiments to other segmentation architectures (including transformer based models), analyzing erosion and dilation independently to quantify their impact, and exploring traditional corruptions such as rotation and translation. Currently, more benchmark experiments are being done on the dataset to detect the presence of any anomalous data points.

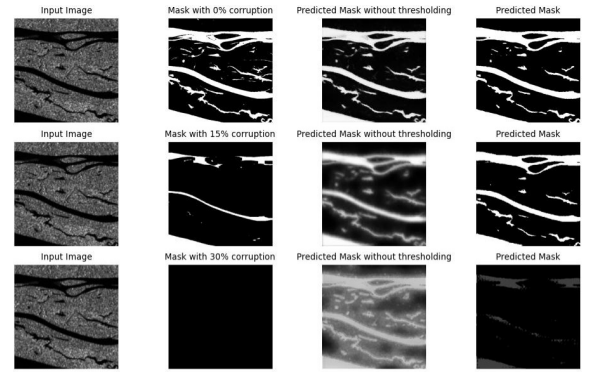


Figure 3: Qualitative results showing the effect of corruption level on the ground truth (second column from the left) and the final predicted (rightmost column) mask, for the input image shown in Figure 1 (leftmost column). The predicted masks before applying the threshold (second column from the right) to the U-Net model outcomes get blurrier with an increase in corruption, but are robust in capturing river structure in the presence of morphological label noise.

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